

AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems

Abstract

Remaining Useful Life (RUL) prediction and estimation within IIoT systems present formidable problems because of the complex degradation process of industrial machinery functioning under different operating conditions and in noise multisensor environments. Although deep learning algorithms have been shown to provide impressive results in prognostics for aero-engines recently, most previous models encounter problems related to their inefficiency in terms of computational complexity, failure to exploit dependencies among different sensors, lack of adaptability to changes in operating conditions, and inability to be deployed effectively in edge-IIoT environments. Moreover, transformer models and other recurrent networks tend to generate high overheads in terms of memory use and latency for long operational sequences of modern industrial systems. Another difficult problem in maintenance systems is accurately estimating long-term degradation while maintaining low energy costs.

In order to overcome these limitations, a new framework titled AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems is introduced. It includes regime-based sensor gating for adaptive modulation, light-weighted temporal compression with stride 1D convolution, graph learning for sensor dependencies modeling, as well as state space selective sequence learning using Mamba for efficient long-term temporal sequence learning. Besides, multitask learning for concurrent Remaining Useful Life estimation and health state classification is employed. Moreover, adaptive edge-aware sensing is used to facilitate an energy-efficient IIoT system deployment. This proposed approach differs from traditional recurrent networks and transformers-based approaches, which require high computational cost despite effective long-term temporal sequence learning capabilities and spatial dependencies representation.

The experimental analysis of the framework was performed using the NASA-N-CMAPSS aero-engine run-to-failure dataset with different operating conditions. The framework performed remarkably well in both regression and classification tasks for predictive maintenance purposes. Specifically, for the regression-based Remaining Useful Life prediction task, the RMSE was 6.3152, the MAE was 4.8296, and the R^2 score was 0.9058, illustrating its ability to estimate the long-term degradation accurately. Moreover, for the health-state classification task, the framework provided a classification accuracy of 84.46%, a macro F1-score of 0.8484, ROC-AUC of 0.9561, and PR-AUC of 0.9195, showcasing its capability to discriminate degradation states and predict critical failures reliably.

Keywords:

Predictive maintenance, Remaining Useful Life prediction, Industrial Internet of Things, aero-engine prognostics, graph neural networks, Mamba, multitask learning.

1. Introduction

Prognostics and Remaining Useful Life (RUL) prediction of aero-engines have emerged as crucial research directions in the era of Industry 4.0 and IIoT owing to the rising need for intelligent maintenance plans and reliable industrial systems that save on energy. Modern aero-engines produce large amounts of multi-sensor time series data across different operational conditions, which makes accurate predictions of their degradation states difficult. Deep learning models for prognostics have proven extremely effective in discovering patterns from raw multi-sensor time series data in recent times. Yet, there remain a number of limitations in existing approaches, including lack of inter-sensor dependencies, adaptation to changing conditions, interpretability issues, inadequate temporal modeling, over-reliance on manual feature engineering, and high computational costs for real-time edge-IIoT deployment.

Some other studies focused on enhancing the accuracy of aero-engine prognostics through the use of graph learning, multi-task learning, attention mechanism, and Spatio-temporal architecture. For example, Ren et al. [1] used a spatial-temporal graph hybrid neural network in aero-engine RUL prediction. This work applied Graph Convolution Networks (GCN) and Graph Transformers together to learn the spatial relationships between sensors and the temporal evolution of degradation. Despite achieving better results, this framework had the problems of high complexity of graphs and transformers, high memory requirements, and large numbers of parameters, which made it hard to deploy on IoT edge devices.

Another example is the study by Hua et al. [2] where they proposed a dynamic feature-aware graph convolution framework for turbofan engine prognostics. In this framework, they employed graph convolution learning to model multisensor interactions and degradation evolution through recurrent temporal learning. The framework showed improvement in terms of learning feature interactions but relied on fixed graph structures and recurrent GRU-based computations, which made it harder to be scalable for long operating time histories.

Wan et al. [3] presented a multitask learning paradigm with health indicators for aero-engine lifetime prediction and health condition evaluation. This study tackled the drawback of single-task learning algorithms for RUL forecasting by conducting RUL prediction and health condition classification in one framework. XGBoost was utilized for engineering health indicators from input data, followed by MMoE-GRU for learning sequential information. Even though satisfactory results were achieved, the reliance on manual feature engineering made this process less flexible and harder to adapt to different datasets. Moreover, recurrent learning added extra computation overhead for large degradation paths.

Kim et al. [4] designed a Spatio-Temporal Koopman Dual-branch Transformer approach for RUL estimation of turbofan engines. In this approach, Koopman operator theory and transformers have been combined to provide an effective way of capturing the behavior of engines despite noise and multi-regime degradation. Even though the method provided good context reasoning and temporal correlation learning, transformer architecture entailed the problem of quadratic complexity, high GPU memory utilization, and high training costs.

Qin et al. [5] introduced an interpretable neurodynamic approach with feature-temporal attention for estimating Remaining Useful Life (RUL). In particular, the introduced approach made use of ODE-based feature augmentation, temporal attention models, and uncertainty-aware prediction methods to enhance interpretability and prediction robustness. Even though the neurodynamic approach enhanced predictability and uncertainty estimation, the extra neurodynamic models posed a higher level of optimization challenges along with the need for tuning attention and uncertainty parameters to ensure stable predictions.

From the present literature review, it is observed that the recent techniques have achieved advancements with respect to graph neural networks, transformers, attention-based approaches, multi-task learning, and explainable artificial intelligence. Nonetheless, most existing techniques exhibit shortcomings such as heavy computational burden, poor energy efficiency, dependency on recurrent neural network architectures, poor scalability in case of long sensor sequences, inadequate adaptability across different operating conditions, and lack of effective spatiotemporal learning capability for IIoT edge applications.

To address these limitations, this work proposes an innovative AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems using a Regime-Gated Graph-Mamba (RG-GMamba) architecture. This architecture uses regime-gating mechanisms for adaptive sensor modulation, light-weight temporal compression by means of stride one-dimensional convolution, graph attention mechanism for modeling sensor dependence, and adaptive state-space sequence learning using Mamba for long horizon degradation analysis. Besides this, the multitask learning framework is applied to simultaneously predict the Remaining Useful Life and health state of the systems.

The main contributions of this work are shown below:

1. A new predictive maintenance framework with regime-aware gating, graph learning, and state-space modeling for aero-engine prognostics using AI techniques.
2. The design of a lightweight Graph-Mamba network that is able to learn spatial sensor dependencies and long-term temporal relationships in an efficient way.
3. Adaptive regime-aware gating methods to dynamically modulate sensor features depending on environmental factors.
4. A multitask learning framework for simultaneous Remaining Useful Life estimation and health-state classification.
5. An energy-efficient edge-IIoT deployment strategy utilizing adaptive sensing and lightweight sequence modeling for real-time industrial monitoring applications.

2. Literature survey

Huang et al. [6] addressed the problem of predicting Remaining Useful Life (RUL) in highly complex industrial systems that exhibit a degradation process at multiple time scales and sensor inter-dependencies. Existing methodologies had only attempted to analyze degradation locally or globally, resulting in the inability to represent a holistic picture of the degradation process.

In order to address this concern, Huang et al. [6] have proposed an evolutionary model called MSEvoNet, which learns the features of degradation at various time scales through a multi-scale evolution framework. This model makes use of multi-scale feature extraction techniques and spatiotemporal representation learning techniques to understand the degradation dependencies at both short and long term scales. Zhang et al. [7] suggested a multi-scale cross-channel attention network for RUL prediction in various sensor configurations. The study found that most of the existing prognostic models considered each sensor channel separately and lacked dynamism when sensor correlation changes occur in different operational modes. The proposed model relied on cross-channel attention to learn relevant inter-sensor relations and suppress irrelevant sensor signals. In addition to cross-channel attention, their model applied multi-scale feature learning to better represent the degradation process in different time scales. Though their model outperformed other models in terms of dynamic learning, it involved heavy computation and needed a large dataset for stable predictions.

The approach of Zhou et al. [8] is called KDMFusion, a knowledge-driven dual-path multi-feature fusion framework for aero-engine RUL prediction. One of the key points emphasized by Zhou et al. [8] was that traditional approaches to feature fusion were prone to fusing feature representations without making proper use of prior degradation information. As a result, such features may turn out to be redundant or sub-optimal. This is why in the proposed approach, dual-path feature extraction together with knowledge-driven fusion schemes was adopted to effectively incorporate prior knowledge about the degradation process. At the same time, the proposed framework provided better opportunities for degradation representation through alignment of interactions between features extracted from multiple information sources. The study by Liu et al. [9] has introduced a two-phase RUL prediction method that utilizes sequence-feature attention techniques. The researchers found out that although previous techniques paid close attention to temporal learning processes, they did not consider any feature interaction amongst various sensor data. The novel framework involved a two-stage learning process wherein the first step was about extracting degradation representations and further improving them by using the sequence-feature attention technique for learning. This helped in focusing on relevant temporal states and sensor data based on which the prediction quality could be enhanced. A dual-path architecture for predicting aero-engine RUL was proposed by Wu et al. [10], based on the technique of time-series decomposition and degradation correction. It was found out by the researchers that existing end-to-end models tend to include not only trends but also non-stationary elements within the sensor sequence. In order to address this issue, the proposed scheme involves a pre-processing step of dividing sequences into two elements – trend and residue, each treated with different learning paths. Additionally, the proposed algorithm includes a degradation correction block that makes final RUL estimation more accurate. Despite achieving a significant improvement in tracking and predicting degradation, the scheme heavily relied on decomposition accuracy and computational efficiency.

On the other hand, Liu et al. [11] studied the problem of RUL estimation when the deterioration process is characterized by time-varying dependencies and interactions of different sensors. It was noted that most RUL models developed till date had a tendency to be either purely

sequential or had neglected important sensor-related features for accurate degradation modeling. As a solution to this problem, Liu et al. [11] suggested a multi-stage RUL prediction model with a sequence-feature attention-based framework. In this approach, first, the authors tried to learn some initial representations of the deterioration process, which were further enhanced with attention-based feature weighting. This allowed the framework to capture the importance of different degradation states and sensor interactions while estimating RUL. Huang et al. [12] introduced MSEvoNet, which is a multi-scale spatiotemporal evolution network for the prediction of Remaining Useful Life in complex industrial systems. The paper pointed out the multi-temporal scale characteristic of the degradation process, which is often overlooked by most conventional deep neural networks because they only focus on either the long-range or short-range aspects of the degradation process. For this reason, the researchers designed a multi-scale spatiotemporal learning framework, which can identify different degradation characteristics in various temporal scales. Through this design, the degradation features in long-range temporal scales were captured better than before. Nonetheless, the model became relatively more complex.

The multi-scale cross-dimension interaction method with adaptive dilated Temporal Convolution Network (TCN) was proposed by Lu et al. [13] for the task of predicting the remaining useful life of aero-engines. In their work, the authors noted that the traditional sequential method was unable to efficiently learn long-term dependencies due to their nature. The proposed approach used adaptive dilated TCN layers to increase the temporal receptive field with minimal increases in the number of parameters while incorporating cross-dimension interaction modules to facilitate interactions among different degradation features. However, this approach is based only on the use of convolutional temporal learning, without inter-sensor topology modelling. Wen et al. [14] presented an adversarial feature decoupling approach for Remaining Useful Life prediction in aero-engine under variable operating regimes. It was found in the work that the change of operational regime often resulted in the phenomenon of domain shift and feature entanglement, leading to the poor generalization performance of traditional prognostic models. In order to address this issue, adversarial feature decoupling was used to disentangle the degradation-related feature from the operational regime-related feature. Additionally, adversarial optimization along with residual learning was incorporated to strengthen the robustness of degradation modeling. Despite the advantages provided by such a method, adversarial optimization posed challenges to the stability of training process. Kapoor et al. [15] developed STARNet, a stacked transfer-aware approach to robustly predict the Remaining Useful Life in multi-regime C-MAPSS engines. The authors noted that many current prognostics algorithms exhibit poor generalization capabilities due to the significant discrepancy in operation conditions between the training and testing phases. In this study, transfer-aware learning components were incorporated into the model development process to enhance the generalization capability by adapting the features across the different operational regimes without losing the sensitive information associated with degradation. Nevertheless, the application of the transfer-learning-based design made the model vulnerable to adaptation challenges.

Li et al. [16] have developed the semi-supervised domain adaptation framework with auxiliary task learning for RUL prediction, wherein the authors pointed out that traditional supervised learning models face problems related to poor generalization due to the scarcity of labeled degradation data in multiple operating domains. They have utilized the concepts of auxiliary task learning and semi-supervised adaptation to achieve improved representations of degradation across various engine conditions. In a related effort, Hatipoğlu and Yılmaz [17] have designed a matrix-statistics-aware attention mechanism for robust aero-engine RUL estimation. The research has taken into consideration the problem related to the inefficacy of traditional attention mechanisms which do not take into account the statistical relationships among the degradation features. In this context, the proposed approach uses matrix-statistics-aware attention learning to develop improved correlations between features while making the model robust to noisy operational conditions. Moreover, Maulana et al. [18] have devised an explainable data-driven framework for RUL prediction in combination with Bayesian Filtering for the NASA CMAPSS datasets. In this regard, the authors observed that many DL-based systems operate as black-box models that offer little insight into their inner working. Berghout et al. [19] proposed a novel deep neural architecture called ProgNet for predicting aircraft engine damage propagation in real flight conditions. In their research, they emphasized the need to develop an approach that enhances the transferability performance for the problem of aircraft engines prognostics under real flight conditions using generalized degradation patterns learned from real flight data. Even though their framework enhanced robustness in real operational conditions, the use of transfer learning added extra complexities, requiring much degradation trajectory learning. Arias Chao et al. [20] developed the official N-CMAPSS run-to-failure dataset under real flight conditions used for prognostics and diagnostics. The research tried to solve the existing gap of degradation datasets in real-world flight conditions in order to enhance aero-engine prognostics.

3. Proposed methodology

3.1 Dataset Description

The proposed model was evaluated using the NASA N-CMAPSS (Nonlinear Commercial Modular Aero-Propulsion System Simulation) dataset, which is a large-scale aero-engine run-to-failure benchmark dataset developed for prognostics and health management research. The dataset includes simulation of realistic turbofan engine degradation patterns under varying flight conditions and operational regimes, making it highly useful for Remaining Useful Life (RUL) prediction and Industrial Internet of Things (IIoT)-based predictive maintenance applications.

The N-CMAPSS dataset is divided into development and testing splits each consisting of several datasets corresponding to operating regimes, physical sensor values, virtual sensors, health metrics, auxiliary inputs, and Remaining Useful Life tags. The complete structure of the dataset is shown in Table 1.

Table 1. N-CMAPSS Dataset Structure

Dataset Group	Variable Name	Description
Development Data	W_{dev}	Scenario descriptors / operational conditions
	X_{s_dev}	Physical sensor measurements
	X_{v_dev}	Virtual sensor measurements
	T_{dev}	Health parameters
	Y_{dev}	Remaining Useful Life (RUL) labels
	A_{dev}	Auxiliary data
Test Data	W_{test}	Scenario descriptors / operational conditions
	X_{s_test}	Physical sensor measurements
	X_{v_test}	Virtual sensor measurements
	T_{test}	Health parameters
	Y_{test}	Remaining Useful Life (RUL) labels
	A_{test}	Auxiliary data
Variable Descriptions	W_{var}	Operational condition variable names
	X_{s_var}	Physical sensor variable names
	X_{v_var}	Virtual sensor variable names
	T_{var}	Health parameter variable names
	A_{var}	Auxiliary variable names

As shown in Table 1, the dataset includes real sensor measurement values along with virtual sensor variable data obtained from simulated systems of aero-engines functioning at different flights conditions. Operational condition variables ((W)) were used for gating regimes of engines, whereas physical sensor measurement ((X_s)) and virtual sensor measurement ((X_v)) values were considered for learning spatial-temporal engine degradation representation using the proposed RG-GMamba approach.

The physical sensor measurement values include the real aero-engine measurement parameters such as fuel flow, fan speed, core speed, temperature and pressure measurements at different levels of compressors and turbines. Furthermore, virtual sensor data include 14 degradation sensitive variables related to flow dynamics, stall margins, and thermodynamics, among other variables. The main physical sensor and virtual sensor variables used in this study are listed in Table 2.

Table 2. Physical and Virtual Sensor Measurements in N-CMAPSS Dataset

No	Sensor Type	Sensor Variable	Description	Unit
1	S	(W_f)	Fuel Flow	pps
2	S	(N_f)	Physical fan speed	rpm
3	S	(N_c)	Physical core speed	rpm
4	S	(T24)	Total temperature at LPC outlet	°R
5	S	(T30)	Total temperature at HPC outlet	°R
6	S	(T48)	Total temperature at HPT outlet	°R
7	S	(T50)	Total temperature at LPT outlet	°R
8	S	(P15)	Total pressure in bypass duct	psia
9	S	(P2)	Total pressure at fan inlet	psia
10	S	(P21)	Total pressure at fan outlet	psia
11	S	(P24)	Total pressure at LPC outlet	psia
12	S	(Ps30)	Static pressure at HPC outlet	psia
13	S	(P40)	Total pressure at burner outlet	psia
14	S	(P50)	Total pressure at LPT outlet	psia
15	VS	(T40)	Burner outlet temperature	°R
16	VS	(P30)	Total pressure at HPC outlet	psia
17	VS	(P45)	Total pressure at HPT outlet	psia
18	VS	(W21)	Fan flow	pps
19	VS	(W22)	Flow out of LPC	lbm/s
20	VS	(W25)	Flow into HPC	lbm/s
21	VS	(W31)	HPT coolant bleed	lbm/s
22	VS	(W32)	HPT coolant bleed	lbm/s
23	VS	(W48)	Flow out of HPT	lbm/s
24	VS	(W50)	Flow out of LPT	lbm/s
25	VS	(SmFan)	Fan stall margin	—
26	VS	(SmLPC)	LPC stall margin	—

27	VS	(SmHPC)	HPC stall margin	—
28	VS	(\Phi)	Fuel-to-pressure ratio	pps/psi

The dataset was divided into training, validation, and testing partitions using engine-wise splitting to prevent trajectory-level data leakage. A sliding window approach was applied to generate samples with temporal degradation information which could be modeled using deep learning based techniques. Around 90,000 samples for training, 13,500 samples for validation, and 27,000 samples for testing purposes were used during the experiments. Standard feature normalization was applied before training in order to enhance optimization efficiency and convergence. The real-world operating conditions and sensor degradation behavior provided by the N-CMAPSS dataset make it extremely suitable to test the proposed energy efficient RG-GMamba based predictive maintenance technique for IIoT applications. The N-CMAPSS dataset used here was taken from the NASA (PCoE) Data Repository.

Dataset Repository: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>

3.2 Data Preprocessing

Data preprocessing was done to enhance the learning stability and predictability of the proposed RG-GMamba framework. Initially, operational condition variables, physical sensor measurements, virtual sensor values, and Remaining Useful Life (RUL) labels were extracted from the NASA N-CMAPSS dataset. Invalid samples and incomplete trajectories were removed to maintain consistency across all dataset subsets. One of the dataset partition files called N-CMAPSS_DS08d-010.h5 could not be read using the HDF5 library and hence caused OSError while processing.

Normalization of features was done by the StandardScaler technique in order to decrease numerical variance among sensors and optimize the stability during training. Because there are multivariate sensor readings with different units and scales within the dataset, normalization helped avoid dominating sensor readings and their negative effect on learning. For temporal degradation learning purposes, sliding window sequence generation was used to turn continuous engine trajectories into fixed-length multivariate time series. Each window sequence had an assigned corresponding RUL value to use for regression learning. At the same time, RUL values were turned into three health states: Normal, Warning, and Critical, in order to perform multi-task classification learning.

Engine-wise data splitting was used to avoid data leakage across training, validation, and testing datasets. Around 90,000 training samples, 13,500 validation samples, and 27,000 testing samples were used during experiment. Finally, sequences and labels were turned into tensor format.

3.3 Proposed Methodology

The proposed AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems uses the Regime-Gated Graph-Mamba (RG-GMamba) architecture which is developed to forecast the remaining useful life and categorize the health state of the engines under different operational regimes. The framework utilizes regime-aware sensor adjustment, temporal feature extraction, graph-based sensor interaction learning, selected state-space modeling, multitask learning, and adaptive edge-aware sensing methods to achieve energy-efficient predictive maintenance in IIoT environments. Multivariate sensor sequences from the NASA N-CMAPSS dataset are provided as input to the framework. The operational condition variables are first processed using a Regime-Aware W-Gate module, which dynamically generates sensor-specific importance weights according to the current engine operating regime. These gate values are multiplied with the raw sensor measurements to suppress noisy or less informative sensor responses while emphasizing degradation-sensitive features.

Gated sensor sequences are then fed into the Strided 1D Convolutional Neural Network (1D-CNN) that extracts temporal features from the sequences. In this step, short-term patterns of degradation are recognized, and temporal redundancies are reduced. The extracted features are further processed by a Graph Attention learning network where each sensor is taken as a graph vertex, and interrelationships between different sensors are automatically learned, considering thermodynamic and mechanical dependencies between sensors. These graph-aided representations are fed into the Mamba selective state-space learning network that learns efficiently the long-term degradation processes. Unlike other models like RNNs and Transformers, Mamba selectively learns long degradation series with linear complexity.

Finally, global temporal pooling is performed, and the resulting representation vectors are passed into two parallel prediction heads that perform multitask learning. One prediction head does regression-based RUL prediction, whereas the other prediction head performs health state classification into Normal, Warning, and Critical states.

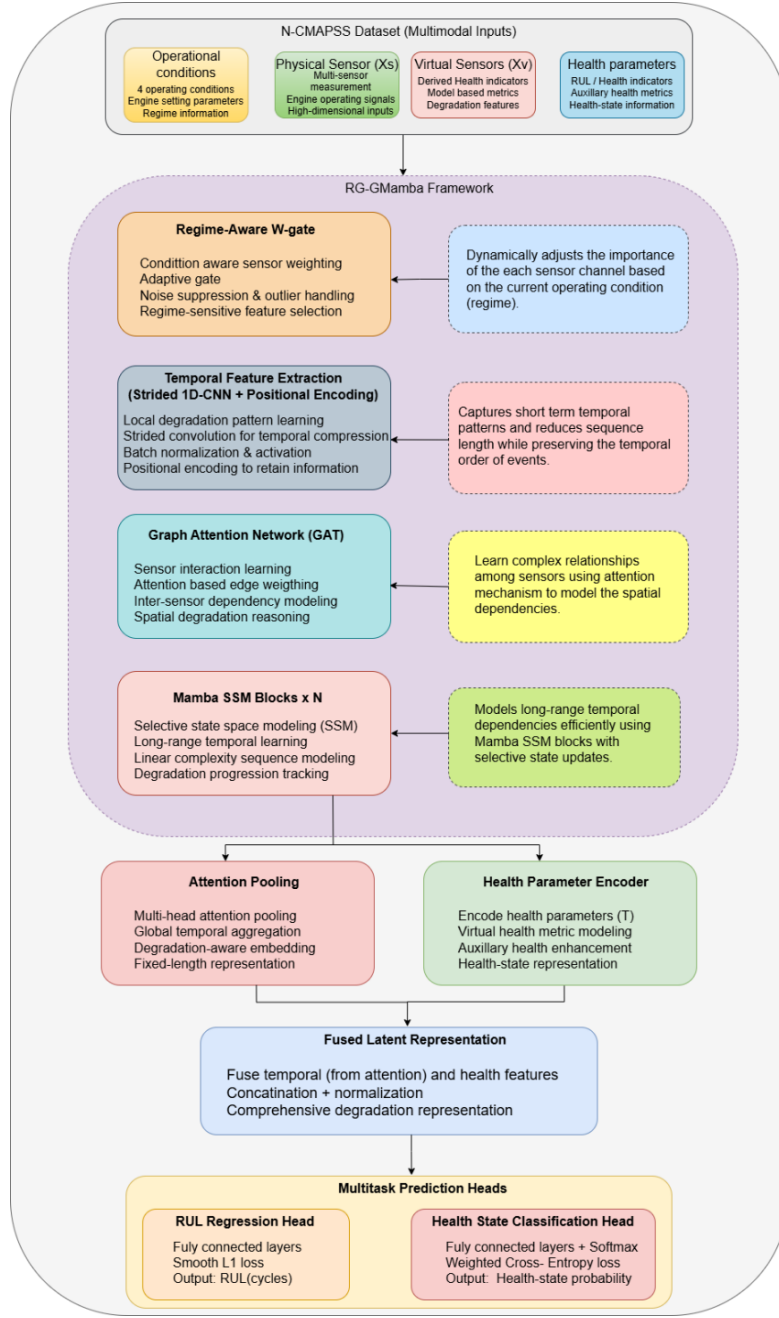


Figure 1. Overview of the proposed RG-GMamba architecture.

3.4 Model Architecture

The proposed architecture is based on a lightweight spatio-temporal prediction maintenance framework called Regime-Gated Graph-Mamba (RG-GMamba), intended for RUL prediction and health-state classification in the realm of IIoT. Contrary to traditional prediction approaches based on recurrent or transformer models, the proposed architecture incorporates regime-aware gating, temporal compression, graph learning of dependencies among sensors, as well as selective state-space sequences modeling to predict equipment degradation effectively with less computation cost and energy consumption. It consists of several modules such as Regime-Aware W-Gate, Strided 1D CNN, Graph Attention Learning, Mamba-based Temporal Modeling, Multi-task Learning, and Adaptive Sensing.

3.4.1 Multivariate Sensor Input Representation

The multivariate sensor sequence is represented as:

$$X \in \mathbb{R}^{B \times C \times L} \quad (1)$$

where:

- B denotes batch size,
- C represents the number of sensor channels,
- L denotes temporal sequence length.

The framework processes:

$$X = \{W, X_s, X_v\} \quad (2)$$

where:

- W represents operational condition variables,
- X_s denotes physical sensor measurements,
- X_v represents virtual sensor variables.

These features include degradation-sensitive data collected from the trajectory patterns of the aircraft engines during varied flight operations. Using this multivariate feature representation, the learning framework can simultaneously capture operational behavior, degradation signatures, and temporal patterns in diverse industrial monitoring systems.

3.4.2 Regime-Aware W-Gate Module

Operating environment affects sensor response in different flight regimes. Hence, the Regime-aware W-Gate module is designed to adjust the weightage accorded to the sensors in the current regime.

The gating mechanism is formulated as:

$$G(t) = \sigma(W_g W(t) + b_g) \quad (3)$$

where:

- $W(t)$ denotes operational condition variables,
- W_g and b_g represent trainable gating parameters,
- σ denotes sigmoid activation,
- $G(t)$ represents sensor-wise gate weights.

The gated sensor sequence is expressed as:

$$\tilde{X}(t) = X(t) \odot G(t) \quad (4)$$

where:

- $\odot$ denotes element-wise multiplication.

This step helps to reduce noise in sensors and enhance the significance of features sensitive to degradation despite operating in different scenarios. The gates help to adaptively weigh the importance of different sensors based on their operation, thus making them robust during volatile flights.

3.4.3 Temporal Feature Extraction using Strided 1D-CNN

The gated sensor sequence is processed using lightweight Strided 1D Convolutional Neural Networks for local temporal feature extraction and sequence compression.

The convolution operation is denoted as:

$$F_{cnn} = \sigma(W_c * \tilde{X} + b_c) \quad (5)$$

where:

- W_c represents convolution kernels,
- $*$ denotes convolution operation,
- b_c denotes bias parameters,
- F_{cnn} represents extracted temporal features.

Batch normalization is applied as:

$$F_{bn} = \frac{F_{cnn} - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (6)$$

where:

- μ and σ^2 denote batch mean and variance,
- ϵ is a stabilization constant.

Strided convolution further minimizes the redundancy in time and computational cost required for future sequential processing. The CNN component helps detect short-term signals of degradation and correlations in time of sensor readings, whereas batch normalization contributes to stabilization of features and faster convergence.

3.4.4 Graph Attention-Based Sensor Interaction Learning

For modeling the dependencies between the sensors, each sensor channel is modeled as a graph node and handled through the Graph Attention based learning approach.

The attention weight for the relationship between sensors is determined by:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [Wh_i \parallel Wh_j]))}{\sum_k \exp(\text{LeakyReLU}(a^T [Wh_i \parallel Wh_k]))} \quad (7)$$

where:

- h_i and h_j denote sensor node representations,
- W represents trainable transformation matrices,
- a denotes attention parameters,
- $\parallel$ represents concatenation.

The updated graph representation is expressed as:

$$h'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} Wh_j \right) \quad (8)$$

This stage helps in achieving dynamic learning of dependencies between aero-engines' sensors from both the thermodynamic and mechanical perspectives. The Graph Attention mechanism adaptively learns the importance of sensor dependencies and enhances spatial degradation reasoning by learning the interactions between sensor channels involving pressure, temperature, vibration, and rotation speed.

3.4.5 Mamba-Based Selective State-Space Modeling

Temporal representations are forwarded into the Mamba-based selective state-space module for long-range degradation learning.

The continuous-time state-space formulation is expressed as:

$$h'(t) = Ah(t) + Bx(t) \quad (9)$$

$$y(t) = Ch(t) \quad (10)$$

where:

- $h(t)$ denotes hidden states,
- $x(t)$ represents input sequences,
- A , B , and C are learnable state-space parameters.

Unlike transformer models, the Mamba module ensures linear computational complexity and learns to represent long sequences of operations as well as their degradation process. This is achieved by using a selective state-space design that adapts the latent representations based on critical degradation occurrences and removes unnecessary time-related data.

3.4.6 Multitask Learning Framework

After temporal modeling, global pooling is applied to create compact degradation-aware feature representations:

$$F_{pool} = Pool(F_{mamba}) \quad (11)$$

The pooled representation is then forwarded into two parallel prediction heads.

The regression-based Remaining Useful Life prediction is expressed as:

$$\hat{Y}_{RUL} = MLP_{reg}(F_{pool}) \quad (12)$$

The health-state classification output is computed as:

$$\hat{Y}_{cls} = Softmax(MLP_{cls}(F_{pool})) \quad (13)$$

where:

- MLP_{reg} denotes regression layers,
- MLP_{cls} represents classification layers.

The framework jointly predicts:

- Remaining Useful Life,
- Normal state,
- Warning state,
- Critical state.

In the multitask learning approach, it is possible to perform representation learning with the consideration of degradation awareness in order to achieve goals relating to prognostics and diagnostics.

3.4.7 Adaptive Edge-Aware Sensing

To achieve energy-efficient IIoT deployment, the process of adaptive sensing is included in the framework. With the use of predicted states of health, the sampling rate can be changed accordingly:

$$f_{sensor} = \begin{cases} f_{low}, & \text{Normal state} \\ f_{high}, & \text{Critical state} \end{cases} \quad (14)$$

where:

- f_{low} denotes low-frequency sensing,
- f_{high} represents high-frequency monitoring.

This strategy reduces energy consumption during stable operation while enabling fine-grained monitoring during critical degradation stages. The adaptive sensing mechanism supports lightweight edge-aware deployment and improves resource utilization efficiency for practical IIoT predictive maintenance environments.

3.4.8 Loss Function and Optimization

The complete RG-GMamba framework is trained using multitask optimization. The overall loss function is defined as:

$$L_{total} = L_{RUL} + \lambda L_{cls} \quad (15)$$

where:

- L_{RUL} denotes regression loss,
- L_{cls} represents weighted cross-entropy classification loss,
- λ is a balancing coefficient.

In multitask optimization, the optimization process optimizes both regression and classification losses simultaneously at the time of training. Balancing factor avoids the dominance of a single optimization criterion and makes multitask learning possible. Such optimization process helps to improve the RUL estimation performance without losing the ability to classify the state of degradation. The proposed architecture is light-weighted, degradation-learning capable, efficient space-time modeling, and energy-efficient predictive maintenance.

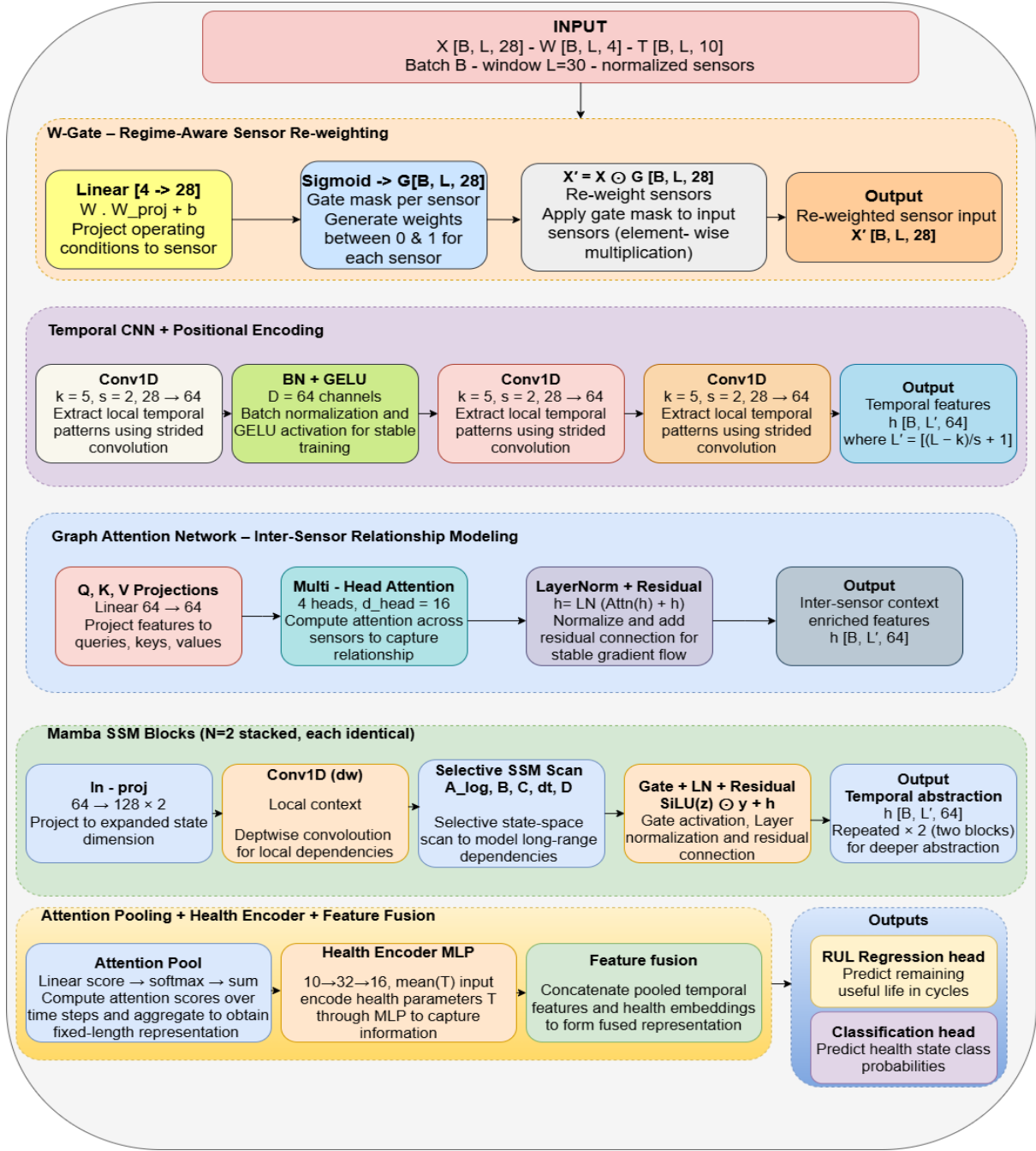


Figure 2. Architecture of the proposed RG-GMamba architecture.

Algorithm

Algorithm 1: RG-GMamba Framework for Energy-Efficient Predictive Maintenance in IIoT Systems

Input:

NASA N-CMAPSS datasets, operational condition variables, multisensor time-series sequences, preprocessing settings, hyperparameters, and health-state labels.

Output:

Trained RG-GMamba predictive maintenance model M , Remaining Useful Life predictions, health-state classification outputs, evaluation metrics, and visualization results.

BEGIN

1: Data Loading and Preparation

- 1.1 Load N-CMAPSS dataset subsets.
- 1.2 Extract operational condition variables W , physical sensors X_s , virtual sensors X_v , and RUL labels Y .
- 1.3 Remove corrupted and unreadable dataset subsets.
- 1.4 Organize engine trajectories into structured multivariate sequences.

2: Dataset Splitting

- 2.1 Split engine trajectories into training, validation, and testing sets.
- 2.2 Ensure engine-wise partitioning to prevent data leakage.
- 2.3 Preserve degradation distribution across all splits.

3: Data Preprocessing

- 3.1 Remove missing records, duplicated samples, and incomplete trajectories.
- 3.2 Apply StandardScaler normalization on sensor features.
- 3.3 Generate fixed-length temporal sequences using sliding-window processing.
- 3.4 Convert RUL values into Normal, Warning, and Critical health states.
- 3.5 Construct PyTorch tensor datasets for GPU-based training.

4: Model Initialization

- 4.1 Initialize the Regime-Aware W-Gate module.
- 4.2 Initialize Strided 1D-CNN temporal compression layers.
- 4.3 Initialize Graph Attention-based sensor interaction layers.
- 4.4 Initialize Mamba selective state-space blocks.
- 4.5 Initialize regression and classification prediction heads.
- 4.6 Initialize multitask loss functions and optimizer parameters.

5: Regime-Aware Sensor Modulation

FOR each input sequence batch (W, X_s, X_v):

- 5.1 Extract operational condition variables W .
 - 5.2 Generate sensor-specific gate weights using the W-Gate module.
 - 5.3 Multiply gate weights with raw sensor sequences.
 - 5.4 Suppress noisy sensor responses and emphasize degradation-sensitive features.
- END FOR

6: Temporal Feature Extraction

- 6.1 Pass gated sequences through Strided 1D-CNN layers.
- 6.2 Extract local temporal degradation patterns.
- 6.3 Reduce sequence length using strided convolution.
- 6.4 Apply batch normalization and GELU activation.

7: Graph-Based Sensor Interaction Learning

- 7.1 Treat each sensor channel as a graph node.
- 7.2 Construct dynamic sensor interaction representations.
- 7.3 Compute graph attention coefficients between sensors.
- 7.4 Aggregate neighboring sensor information using attention learning.
- 7.5 Generate graph-enhanced degradation features.

8: Mamba-Based Temporal Sequence Modeling

- 8.1 Forward graph-enhanced sequences into Mamba blocks.
- 8.2 Perform selective state-space sequence learning.
- 8.3 Capture long-range degradation dependencies efficiently.
- 8.4 Suppress redundant temporal information using selective memory updates.

9: Global Representation Learning

- 9.1 Apply global temporal pooling on Mamba outputs.
- 9.2 Generate compact degradation-aware feature representations.

10: Multitask Prediction Framework

- 10.1 Forward pooled features into regression prediction head.
- 10.2 Predict Remaining Useful Life values.
- 10.3 Forward pooled features into classification prediction head.
- 10.4 Predict Normal, Warning, and Critical health states.

11: Loss Construction

- 11.1 Compute regression loss for RUL prediction.
- 11.2 Compute weighted cross-entropy loss for health-state classification.
- 11.3 Apply class weights to address imbalance.
- 11.4 Combine regression and classification losses into multitask loss.

12: Training Loop

FOR each epoch DO

- 12.1 Load training batches.
- 12.2 Perform forward propagation through RG-GMamba.
- 12.3 Compute multitask loss.
- 12.4 Backpropagate gradients.
- 12.5 Clip gradients for stable optimization.
- 12.6 Update parameters using AdamW optimizer.
- 12.7 Evaluate performance on validation data.
- 12.8 Save best model checkpoint if validation loss improves.
- 12.9 Stop training early if validation performance stagnates.

END FOR

13: Model Evaluation

- 13.1 Restore the best saved model.
- 13.2 Evaluate regression performance using RMSE, MAE, R^2 , and NASA score.
- 13.3 Evaluate classification performance using Accuracy, F1-score, ROC-AUC, and PR-AUC.
- 13.4 Compute per-dataset performance metrics for all N-CMAPSS subsets.

14: Visualization and Explainability

- 14.1 Generate training and validation loss curves.
- 14.2 Generate confusion matrix and ROC curves.
- 14.3 Generate sensor attention and gate activation visualizations.
- 14.4 Generate SHAP and LIME-based interpretability plots.
- 14.5 Generate degradation trajectory and uncertainty visualizations.

15: Output Results

- 15.1 Return trained RG-GMamba model M .
- 15.2 Save model checkpoints and evaluation metrics.
- 15.3 Save generated visualizations and performance reports.
- 15.4 Export final predictive maintenance results for IIoT deployment.

END

3.5 Loss Functions

In the proposed framework known as RG-GMamba, multitask optimization is used for the two tasks of predicting the RUL and classifying the state of health. Given that both the regression and classification tasks are part of the same framework, different loss functions will be used for the tasks and integrated into an optimization objective function.

In the regression problem for RUL prediction, the MAE loss function is used to reduce the absolute error between the predicted and actual Remaining Useful Life.

The MAE regression loss is defined as:

$$L_{RUL} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (16)$$

where:

- y_i denotes actual RUL values,
- $\hat{y}_i$ represents predicted RUL values,
- N denotes the number of training samples.

The increased MAE loss makes the model more resilient to outliers and makes stable learning of the degradation process in the long run.

For the task of classifying the health state, we use the Softmax activation function to output probabilities for Normal, Warning, and Critical operation conditions:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (17)$$

where:

- z_i denotes classification logits,
- C represents the total number of classes.

To address class imbalance among degradation states, weighted cross-entropy loss is employed and expressed as:

$$L_{cls} = - \sum_{i=1}^c w_i y_i \log(\hat{y}_i) \quad (18)$$

where:

- w_i denotes class weights,
- y_i represents ground-truth labels,
- $\hat{y}_i$ denotes predicted class probabilities.

The final multitask optimization objective combines regression and classification losses as:

$$L_{total} = L_{RUL} + \lambda L_{cls} \quad (19)$$

where:

- L_{RUL} denotes MAE regression loss,
- L_{cls} represents weighted classification loss,
- λ is the balancing coefficient controlling task contribution during optimization.

3.6 Optimization Strategy

The proposed RG-GMamba approach makes use of a multitasking optimization technique that can help ensure a stable convergence, degrade learning efficiency, and health state classifications despite changing operating conditions. AdamW optimization is implemented within this approach because of its ability to provide better weight regularizations and stable optimizations in the case of deep neural models. Gradient clipping is introduced in order to avoid the problem of exploding gradients in cases of long-sequence learning, while early stopping is introduced to avoid overfitting problems. Also, weighted classification learning is implemented for handling imbalances in Normal, Warning, and Critical degradation states. The important optimization configurations for training the proposed approach is shown in Table 3.

Table 3. Optimization Parameters Used for RG-GMamba Training

Parameter	Value
Optimizer	AdamW
Learning Rate	0.001
Weight Decay	0.0001
Batch Size	128
Epochs	60
Early Stopping Patience	12

Gradient Clipping	Enabled
Gradient Clipping Threshold	1.0
Activation Function	GELU
Regression Loss	MAE Loss
Classification Loss	Weighted Cross-Entropy
Scheduler	Cosine Annealing

As shown in Table 3, AdamW optimization with weight decay regularization was used to improve parameter generalization and reduce overfitting during multitask learning. The parameter update rule for AdamW optimization is expressed as:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta \lambda_w \theta_t \quad (20)$$

where:

- θ_t denotes model parameters,
- η represents learning rate,
- $\hat{m}_t$ and $\hat{v}_t$ denote bias-corrected moment estimates,
- λ_w represents weight decay regularization.

To stabilize optimization during long-sequence degradation learning, gradient clipping was applied as:

$$g \leftarrow g \cdot \min\left(1, \frac{\tau}{\|g\|}\right) \quad (21)$$

where:

- g denotes gradients,
- τ represents the clipping threshold.

The learning rate scheduling process was controlled by using cosine annealing for better convergence stability in training:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos\left(\frac{T_{cur}}{T_{max}}\pi\right)\right) \quad (22)$$

where:

- η_t denotes current learning rate,
- η_{max} and η_{min} represent maximum and minimum learning rates,
- T_{cur} denotes current iteration,

- T_{max} represents total training iterations.

The overall optimization approach helped achieve stable convergence for multitasking, avoid overfitting, improve the ability to learn degenerate learning, and ensure effective training for the energy-efficient RG-GMamba predictive maintenance system.

4. Results and Discussion

4.1 Experimental Setup

Multiple N-CMAPSS aero-engine run-to-failure dataset with varying flight and deterioration scenarios have been used. The full predictive maintenance pipeline was created through the use of multivariate operational condition parameters, actual sensor values, virtual sensor parameters, and Remaining Useful Life labels available in the NASA N-CMAPSS dataset. Approximately 90,000 training samples, 13,500 validation samples, and 27,000 testing samples were created by constructing sequences using sliding windows technique based on several engine deterioration trajectories.

The pre-processing pipeline involved removal of corrupted and incomplete trajectories, normalization of features via the StandardScaler method, creation of time-series data, and labeling of the health state of each trajectory into Normal, Warning, and Critical degradation conditions. Operational condition variables were separately kept to modulate sensors according to the regime-based approach in the proposed RG-GMamba architecture. PyTorch framework was used to build the model. Training of the model took place in the Google Colab Pro environment with a GPU of NVIDIA L4 type. Some training techniques such as gradient clipping, cosine annealing, early stopping, and multitask weighted loss were also used.

4.2 Performance Evaluation Metrics

The performance of the proposed RG-GMamba approach was evaluated using both regression and classification metrics to evaluate the accuracy in Remaining Useful Life prediction and health-state classification ability. Regression metrics were used to analyze the accuracy in estimating the degradation, while the classification metrics were utilized for analyzing the ability to discriminate the health states depending on different operating conditions.

Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) measures the square-root average prediction error between actual and predicted Remaining Useful Life values. RMSE penalizes the greater prediction errors. Lower RMSE values shows better prediction accuracy.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (23)$$

where:

- y_i denotes actual RUL values,

- $\hat{y}_i$ represents predicted RUL values,
- N denotes the number of samples.

Mean Absolute Error (MAE)

Mean Absolute Error (MAE) calculates the average absolute difference between actual and predicted Remaining Useful Life values. Compared to Root Mean Square Error (RMSE), MAE is less sensitive to outliers.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (24)$$

where:

- y_i denotes actual RUL values,
- $\hat{y}_i$ represents predicted RUL values.

Coefficient of Determination (R^2)

The coefficient of determination (R^2) evaluates how well the regression model explains degradation variability in the dataset. A high R^2 value means that there is better correlation between predicted and actual RUL values.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (25)$$

where:

- y_i denotes actual RUL values,
- $\hat{y}_i$ predicted RUL values,
- $\bar{y}$ denotes mean RUL value.

Classification Accuracy

Classification accuracy is the percentage of the correctly classified health state cases. High accuracy means good classification abilities under the Normal, Warning, and Critical modes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (26)$$

Precision

Precision evaluates the accuracy rate of predicting positive degradation statuses. The higher the Precision, the less often there will be false alarms in the predictive maintenance system.

$$Precision = \frac{TP}{TP + FP} \quad (27)$$

Recall

Recall measures the ability of the machine learning model to identify degradation statuses accurately. The importance of Recall in the predictive maintenance system comes from the risk of missing failures that can lead to significant issues.

$$Recall = \frac{TP}{TP + FN} \quad (28)$$

F1-Score

F1-score provides a balanced evaluation between Precision and Recall. It is useful for imbalanced degradation datasets where some health states contain fewer samples than others.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (29)$$

Receiver Operating Characteristic Area Under Curve (ROC-AUC)

ROC-AUC evaluates effectiveness of the classifier in classifying different degradation classes at multiple thresholds. The higher the ROC-AUC value, the greater the capability of the algorithm to distinguish between classes.

$$ROC-AUC = \int_0^1 TPR(FPR^{-1}(x)) dx \quad (30)$$

Precision-Recall Area Under Curve (PR-AUC)

PR-AUC evaluates classification efficiency in imbalanced datasets by assessing the correlation between Precision and Recall using multiple thresholds. The PR-AUC criterion is very relevant for the problem of predictive maintenance since critical classes of degradation are typically under-represented.

$$PR-AUC = \int_0^1 Precision(Recall^{-1}(x)) dx \quad (31)$$

Higher PR-AUC values indicate better degradation-state detection capability.

4.3 Performance Analysis and Results

The performance of the proposed RG-GMamba framework was evaluated by considering the performance measures for both regression and classification tasks to assess the prediction of remaining useful life and performance of classification of health state under different operating conditions. Regression measures were considered for assessing the degradation prediction performance, while classification measures were considered to evaluate the degradation-state

discrimination performance. The overall regression performance measure of the proposed RG-GMamba is shown in Table 4, classification performance results is shown in Table 5.

Table 4. Regression Performance of Proposed RG-GMamba Framework

Metric	Value
RMSE	6.3152
MAE	4.8296
MSE	39.8813
R ² Score	0.9058
NASA Score	17630.0938

Based on the results shown in Table 4, the introduced RG-GMamba approach was found to be very effective in predicting the Remaining Useful Life of engines. For instance, the approach generated an RMSE of 6.3152 and an MAE of 4.8296, which is indicative of good degradation predictions and minimization of errors in prediction regardless of changing environmental situations. Additionally, the approach was found to have an R² of 0.9058, which is quite good considering the correlation that existed between the predicted and actual RUL values.

Table 5. Classification Performance of Proposed RG-GMamba Framework

Metric	Value
Accuracy	0.8446
F1 Macro	0.8484
F1 Micro	0.8446
F1 Weighted	0.8434
Precision (Macro)	0.8448
Recall (Macro)	0.8533
ROC-AUC	0.9561
PR-AUC	0.9195
PR-AUC Normal	0.8799
PR-AUC Warning	0.8958
PR-AUC Critical	0.9828

The results for the performance of classification as shown in Table 5 show that the proposed framework provided robust ability to discriminate between different health-states across the Normal, Warning, and Critical degradation states. The framework was able to get an overall classification accuracy of 84.46% and a macro F1-score of 0.8484. This shows that there is balanced performance in the classification across different degradation states. Moreover, the proposed framework was able to obtain very high ROC-AUC and PR-AUC scores of 0.9561 and 0.9195 respectively. The critical-state PR-AUC score was 0.9828.

4.4 Dataset-wise Performance Analysis

Performance analysis of the proposed RG-GMamba algorithm was performed at the dataset level in order to study the ability of RUL prediction and health state classification under various operating regimes. Performance evaluation of the proposed model was carried out based on regression performance measures like RMSE and MAE, as well as the accuracy of health state classification. Dataset level performance results for both regression and classification problems have been visualized in Figure. 3 and quantitative results have been shown in Table 6.

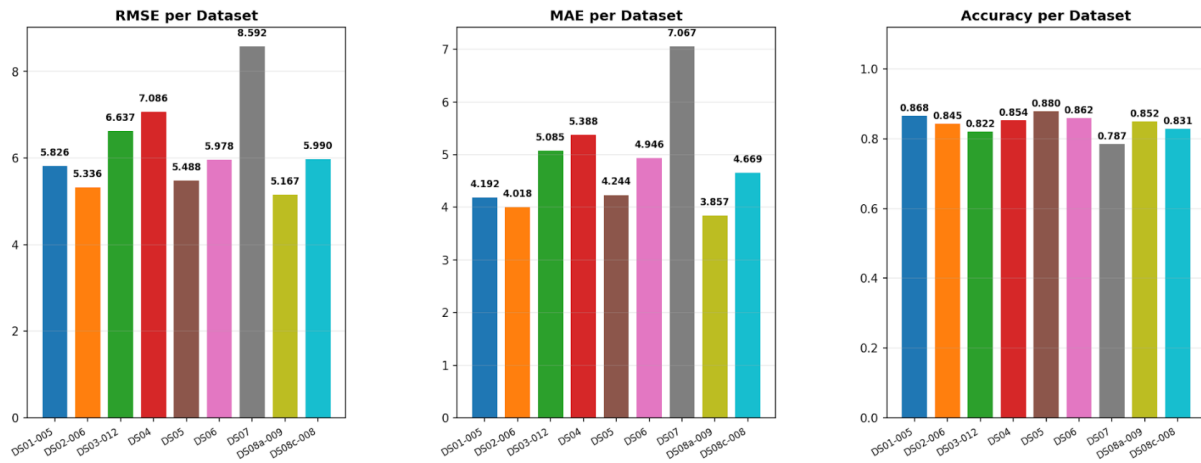


Figure 3. RMSE, MAE, and Accuracy per Dataset Visualization

As shown in Figure. 3 and Table 6, the proposed RG-GMamba framework achieved stable Remaining Useful Life prediction and health-state classification performance across multiple N-CMAPSS dataset subsets. The framework obtained lower RMSE and MAE values for DS08a-009, showing accurate degradation estimation and reduced prediction deviation under varying operational conditions. In DS07 it produced higher RMSE and MAE values due to increased degradation complexity and operational variability present within that dataset subset.

Table 6. Dataset-wise Performance Analysis of Proposed RG-GMamba Framework

Dataset	RMSE	MAE	Accuracy
DS01-005	5.826	4.192	0.868
DS02-006	5.336	4.018	0.845
DS03-012	6.637	5.085	0.822

DS04	7.086	5.388	0.854
DS05	5.488	4.244	0.880
DS06	5.978	4.946	0.862
DS07	8.592	7.067	0.787
DS08a-009	5.167	3.857	0.852
DS08c-008	5.990	4.669	0.831

The classification accuracy analysis clearly shows high discriminative ability on different operation statuses of the aero-engines. The highest classification accuracy value is found to be 0.880 for DS05, whereas the lowest accuracy is attained by DS07, which amounts to 0.787, due to higher complexity in its degradation characteristics. The framework shows excellent performance consistency through all datasets, thus showing generalizability in the real-time predictive maintenance system using IIoT systems.

In summary, from the analysis performed on the entire dataset, it is evident that the developed RG-GMamba framework successfully learns ST degradation patterns efficiently.

4.4 Training Convergence Analysis

The training performance of the RG-GMamba method has been evaluated based on the loss convergence and prediction errors trends through training. This evaluation was intended for assessing the optimization, learning effectiveness, and the generalization capacity of the model during multitasking training. The loss curves for training and validation are shown in Figure. 4, and those of MAE and validation RUL loss are displayed in Figure. 5.

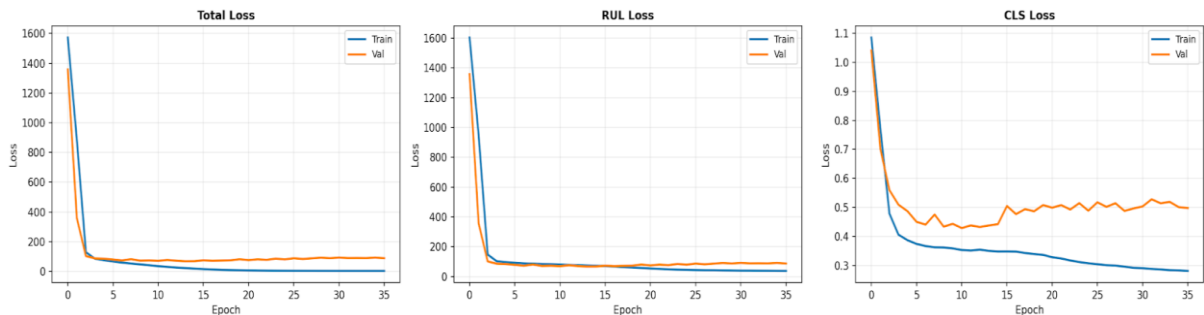


Figure. 4. Training and validation convergence curves of the proposed RG-GMamba framework showing Total Loss, RUL Loss, and Classification Loss across training epochs.

As shown in Figure. 4, both training and validation losses reduce sharply during the first few epochs, showing efficient feature representation learning and optimization process. Both the Total Loss and the RUL Loss converge smoothly and eventually stabilize after the training process proceeds, confirming that effective degradation learning has been achieved through our optimization method. Meanwhile, a clear reduction pattern of the Classification Loss can be observed from its curve. In contrast to the Total and RUL Losses, a slight deviation between the training and validation losses occurs after several epochs during training.

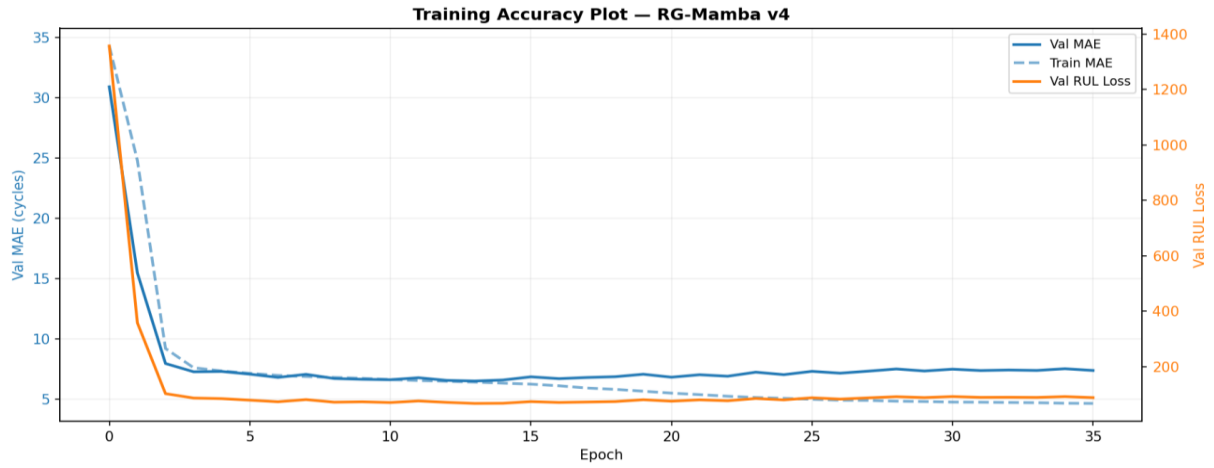


Figure. 5. Training performance analysis of the proposed RG-GMamba framework showing Train MAE, Validation MAE, and Validation RUL Loss across training epochs.

Figure. 5 shows how prediction error evolves during the process of training the model. There is a drop in training and validation MAE at the initial stage, which means fast learning of deterioration patterns based on the N-CMAPSS data set. Validation RUL loss demonstrates the same pattern of evolving where it decreases fast during the early stages of training and converges gradually after. It can be concluded that there is no significant difference between training and validation errors which means low probability of overfitting and high generalization capacity.

4.5 Confusion matrix

Figure. 6 presents the confusion matrix of the proposed RG-GMamba framework for health-state classification. The results shows strong discrimination capability across the three degradation categories, namely Normal, Warning, and Critical.

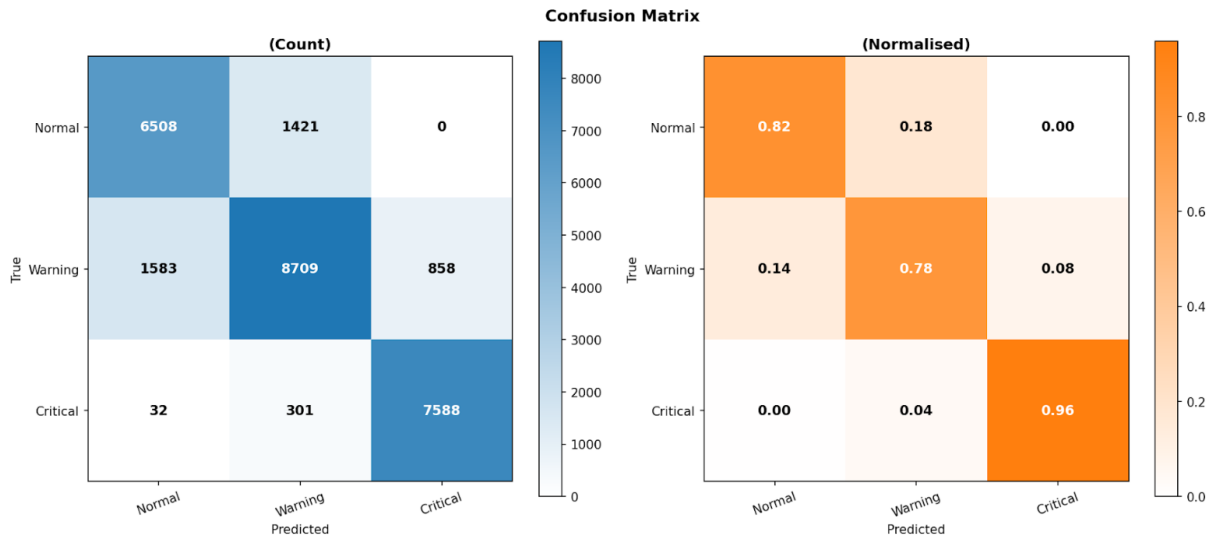


Figure. 6. Confusion matrix of the proposed RG-GMamba framework.

For the Normal class, 6,508 samples were correctly classified, while 1,421 samples were misclassified as Warning. The model achieved a normalized classification rate of 82% for this

class. For the Warning class, a total of 8,709 samples were recognized properly, and 1,583 samples were recognized as Normal and 858 samples as Critical, obtaining the normalized recognition rate of 78%. As per the best recognition rate among others, Critical class had 7,588 samples classified accurately, with 301 recognized as Warning and only 32 recognized as Normal, yielding the normalized recognition rate of 96%.

Normalized confusion matrix shows that just 4% of critical samples have been wrongly classified as Warning and nearly 1% samples as Normal. Likewise, nearly 18% of Normal samples have been recognized as Warning, whereas 14% of Warning samples have been recognized as Normal. In general, misclassifications have mostly taken place between adjacent degradation classes as expected, because of gradual degradation progression.

Figure. 6 demonstrates the confusion matrix that clearly shows that our proposed RG-GMamba framework successfully detects degradation in critical state while maintaining a balanced level of performance in recognition rates of all health states. This is in line with our calculated PR-AUC of 0.9828, which clearly demonstrates the applicability of this framework in the IIoT environment.

4.6 ROC-AUC and Precision–Recall Curves Analysis

Figure. 7 shows the performance of the suggested RG-GMamba model in ROC and PR curves for multiclass health state identification. It can be seen that the ROC curves for the One-vs-Rest case indicate good class separation for all health classes. The performance metrics are ROC-AUC scores of 0.9539 for the Normal class, 0.9216 for the Warning class, and 0.9927 for the Critical class. An outstanding ROC-AUC score for the Critical class implies that the discrimination between health classes is very good.

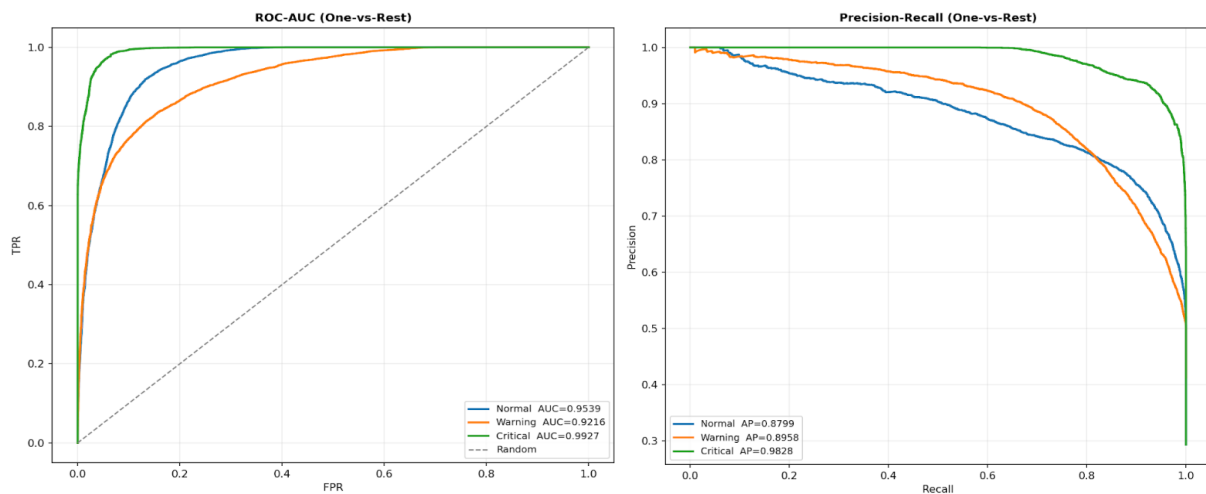


Figure. 7. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves of the proposed RG-GMamba framework.

The Precision-Recall graphs demonstrate the classification reliability even when there is a class imbalance. The Average Precision (AP) values obtained by using the introduced framework for the Normal class, the Warning class, and the Critical class equal 0.8799, 0.8958, and 0.9828,

respectively. High precision at a broad range of recall for the Critical class indicates the ability of the framework to classify objects in such a way that ensures the detection of any failure.

All ROC and PR curves shown in Figure. 7 explains the ability of the proposed RG-GMamba framework to classify states with great accuracy. This is especially important in the case of predictive maintenance where successful classification of the Critical state can directly lead to failure prevention.

4.7 Prediction Confidence and Model Calibration Analysis

In order to further analyze the reliability of the proposed RG-GMamba architecture, analysis of prediction confidence distribution and probability calibration was done. The prediction confidence distribution helps understand the reliability of predictions from the model, as well as help verify whether the predicted probabilities indicate the actual chance of the particular state of health. The prediction confidence distribution is shown in Figure. 8, while calibration reliability charts are given in Figure. 9.

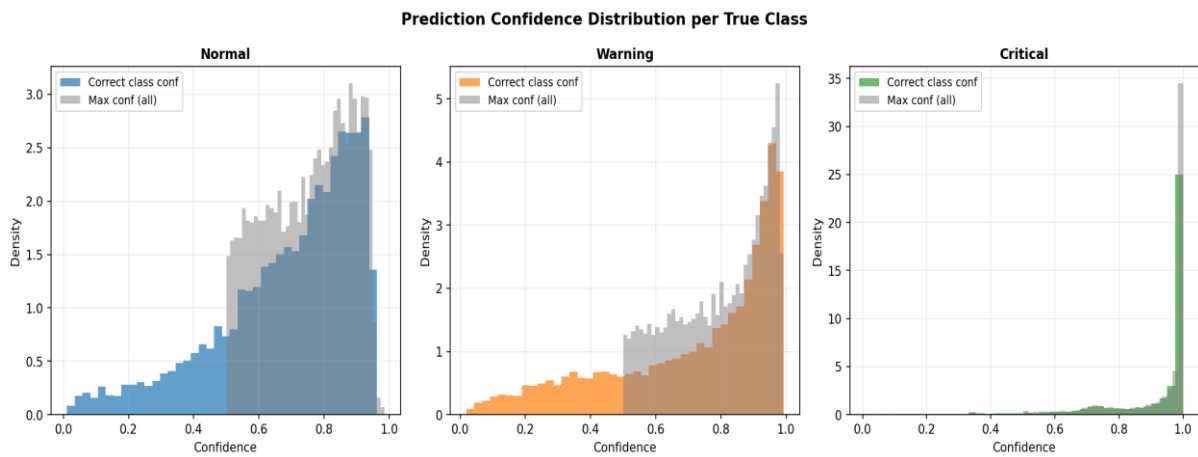


Figure. 8. Prediction confidence distribution of the proposed RG-GMamba framework for Normal, Warning, and Critical health states.

As can be seen in Figure. 8, the prediction distributions for the three health classes have most of their predictions concentrated on the high confidence area at > 0.8 . The predictions for the Critical class are found to be more concentrated around 1.0 which reflects that the proposed method is able to detect the critical condition of failure confidently. In contrast, the distribution for the Warning class is less concentrated because of its intermediary role between normal and critical health conditions. The same case is observed for the Normal class too.

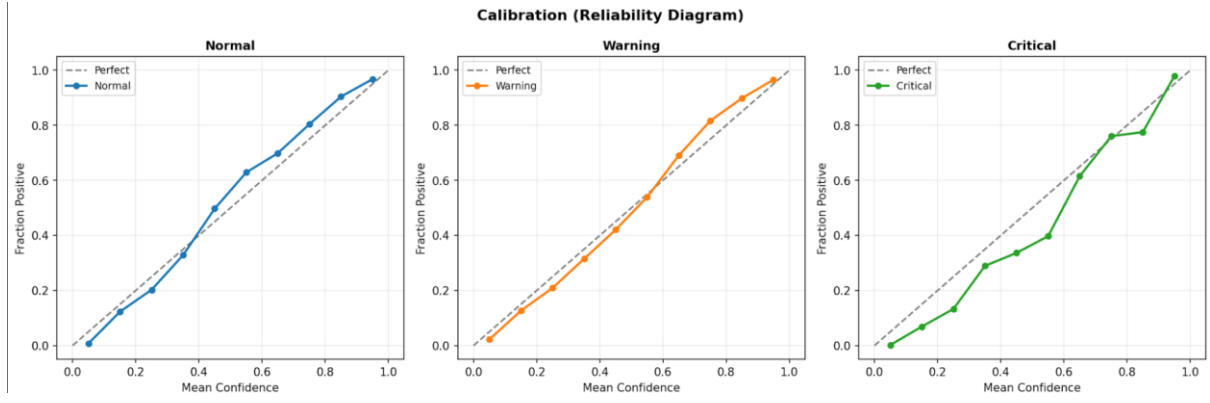


Figure. 9. Reliability diagrams of the proposed RG-GMamba framework showing probability calibration performance for Normal, Warning, and Critical health-state classifications.

Figure. 9 is showing the results of the proposed method's calibration performance using reliability diagrams. It can be seen from Figure. 9 that the curves for the Normal and Warning categories almost coincide with the ideal line, which indicates that there is a strong match between the probability predictions and real outcomes. Despite some deviations occurring in the Critical category in terms of middle confidence values, the general trend still resembles the ideal calibration line, especially in the case of high confidence levels.

4.8 Uncertainty and Failure Forecast Analysis

In order to evaluate the reliability and robustness of the developed RG-GMamba model, experiments on uncertainty estimation, engine level trajectory analysis, and failure prediction were conducted. The findings from these tests can be used to understand the confidence of prediction, tracking of degradation processes, and failure warning ability of the model under different operating scenarios. The results of uncertainty estimation experiment have been shown in Figure 10., while engine-level trajectory and failure forecasting analyses are presented in Figures 11, 12, and 13.

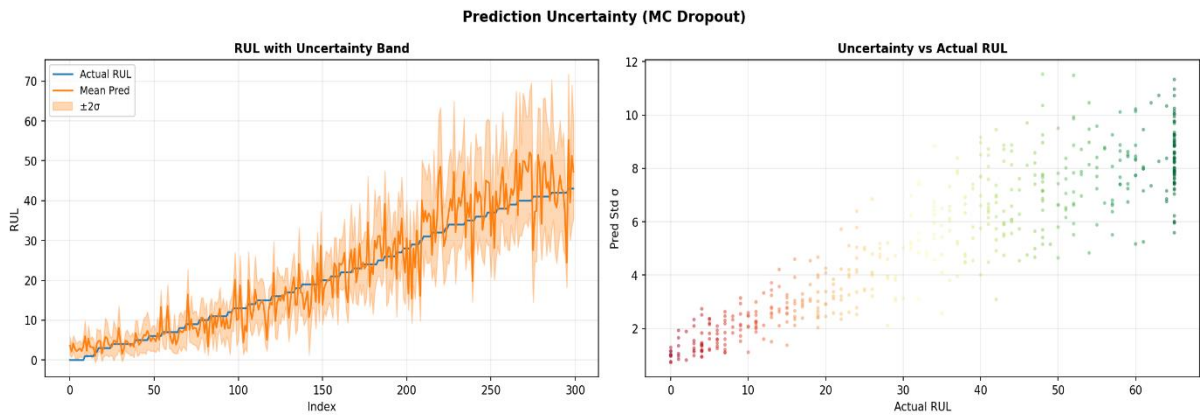


Figure 10. Monte Carlo Dropout-based prediction uncertainty analysis on proposed model.

Figure 10 presents the performance results of the proposed method by depicting RUL prediction with adaptive uncertainty bounds. From the figure 10, it is known that the mean of the prediction curve fits well with the actual RUL, suggesting an accurate tracking capability of the degradation process as operation proceeds. In addition, the uncertainty bound is

relatively small at initial stages of the operation where the operating condition is constant, but becomes wider for larger RUL values. Further, from the uncertainty scatter plot, an increase in prediction variance is observed when degradation becomes complex.

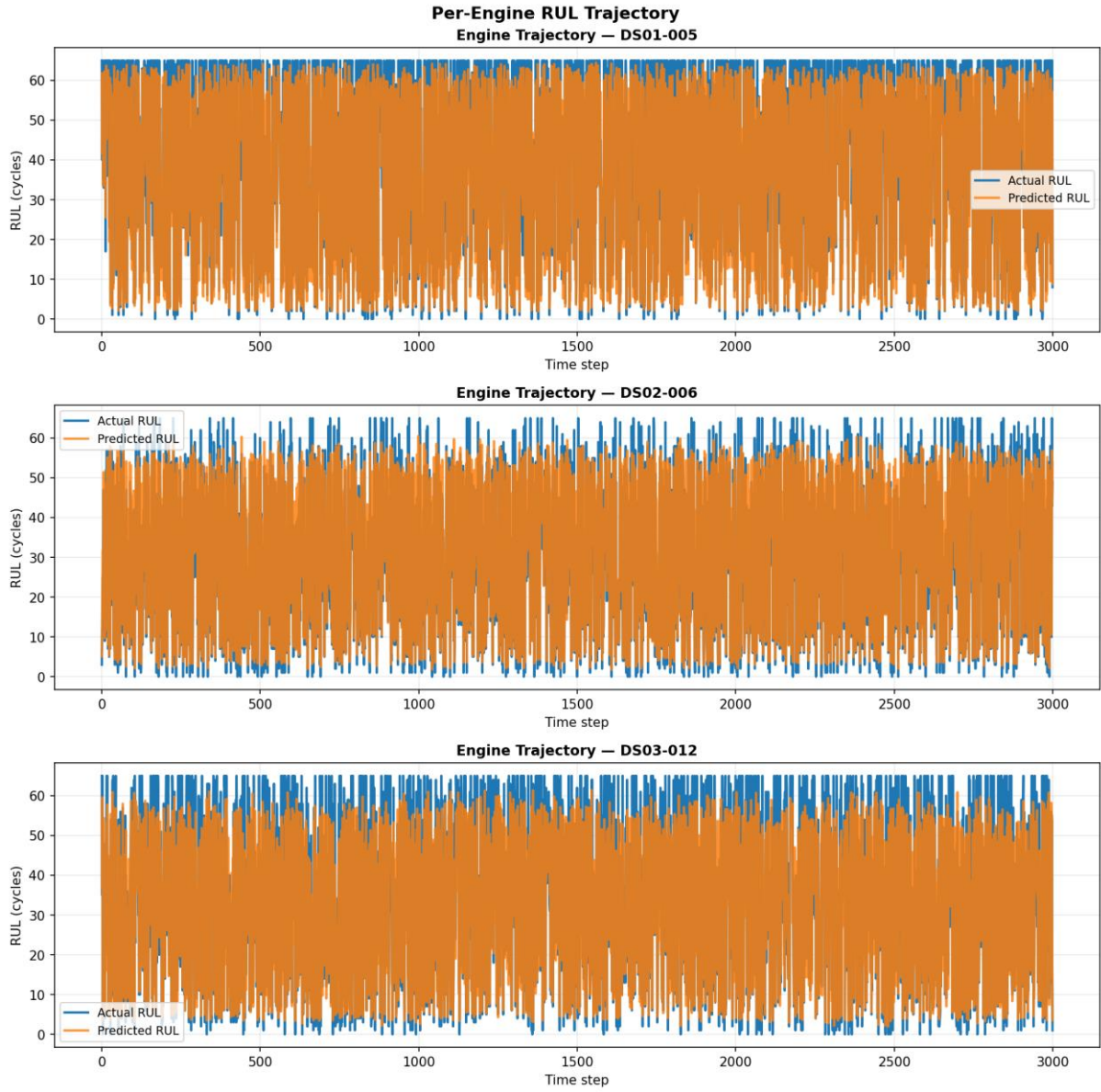


Figure 11. Actual and predicted Remaining Useful Life trajectories across multiple engine degradation sequences.

The engine level analysis of RUL trajectory for multiple datasets from the N-CMAPSS is shown in Figure 11. The predicted RUL trajectory agrees very well with the actual RUL trajectory despite changing environments. For most of the degradation phases, the new approach, RG-GMamba, is able to predict the RUL trajectory very accurately while successfully modeling non-linear variations during the process of degradation. However, small prediction errors can be noticed in highly dynamic phases of degradation but the prediction trend still persists over a period of time.

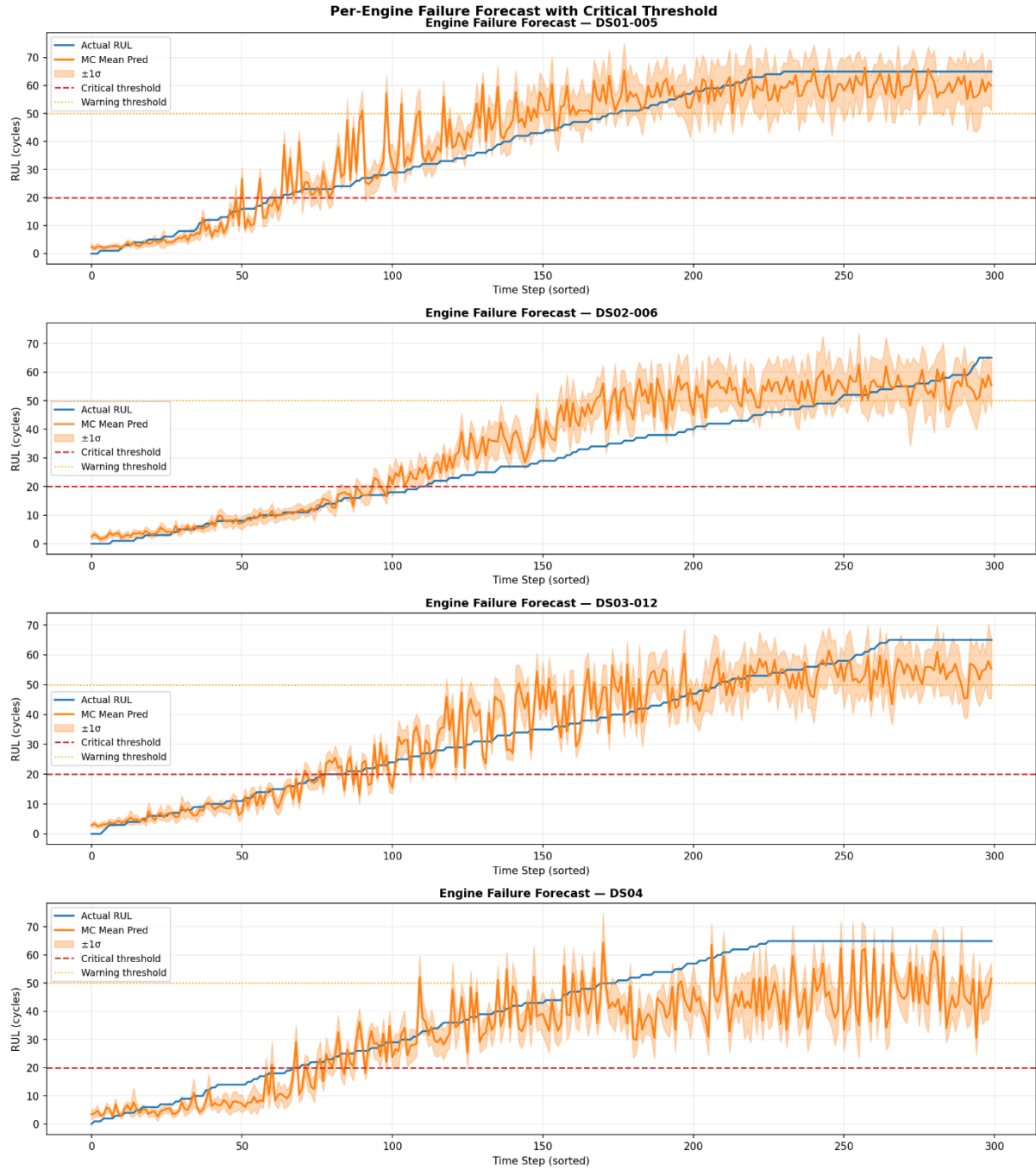


Figure 12. Per-engine failure forecasting with uncertainty bands and operational thresholds.

Figure 12, shows the engine-level failure forecasting behavior with Monte Carlo uncertainty analysis together with Warning and Critical operation limits. The forecasts remain close to the true degradation dynamics while retaining uncertainty estimates adaptive to the forecast average values. The Warning and Critical threshold visualization shows that the developed approach is able to detect risky degradation areas ahead of the severe degradation development phase.

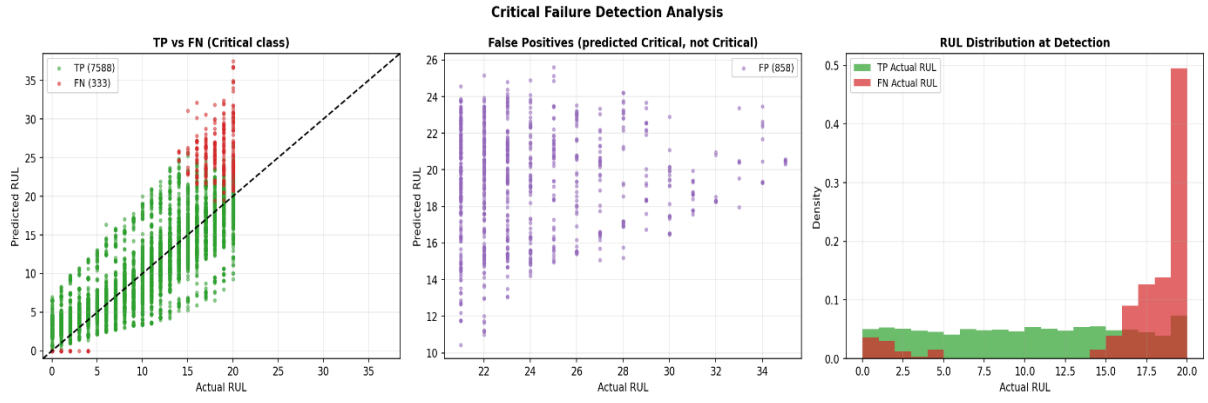


Figure 13. Critical failure detection analysis of the proposed RG-GMamba framework.

Figure 13 shows the important failure detection ability of the suggested approach using True Positive, False Positive, and RUL Distribution Analysis. The performance of the suggested model was very good regarding True Positive behavior, and few false positives were detected when there was serious degradation in the system. The estimated value of the remaining useful life is quite near to the actual value of the critical samples, which means that it predicts the impending state of failure. Also, the failure distribution analysis has shown that the proposed framework successfully detects risky states of operation before any failure occurred.

4.9 Latent Feature Space Visualization Analysis

In order to evaluate the degradation modeling ability of the introduced RG-GMamba approach, the visualization of the latent features was carried out using the UMAP and t-SNE methods. The visualization of the features by means of UMAP is shown in Figure 14, whereas the t-SNE latent space is demonstrated in Figure 15.

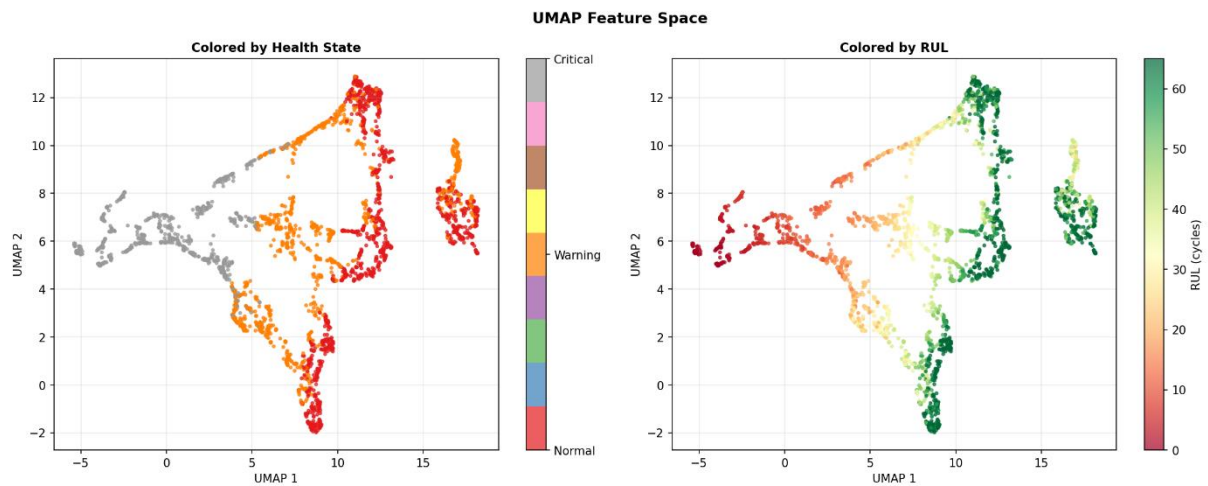


Figure 14. UMAP-based latent feature space visualization of the proposed RG-GMamba framework.

As can be seen from Figure 14, the visualization using UMAP indicates clear clustering properties in terms of the Normal, Warning, and Critical types of degradation states. Latent embeddings present a smooth transition of degradation between operational states, suggesting that the presented approach has successfully modeled the degradation-aware feature

representations from multivariate sequential data. As can also be seen in Figure 14, the visualization using RUL coloring indicates the presence of smooth progression of Remaining Useful Life in the latent space. Samples with similar degradation patterns are clustered nearby each other.

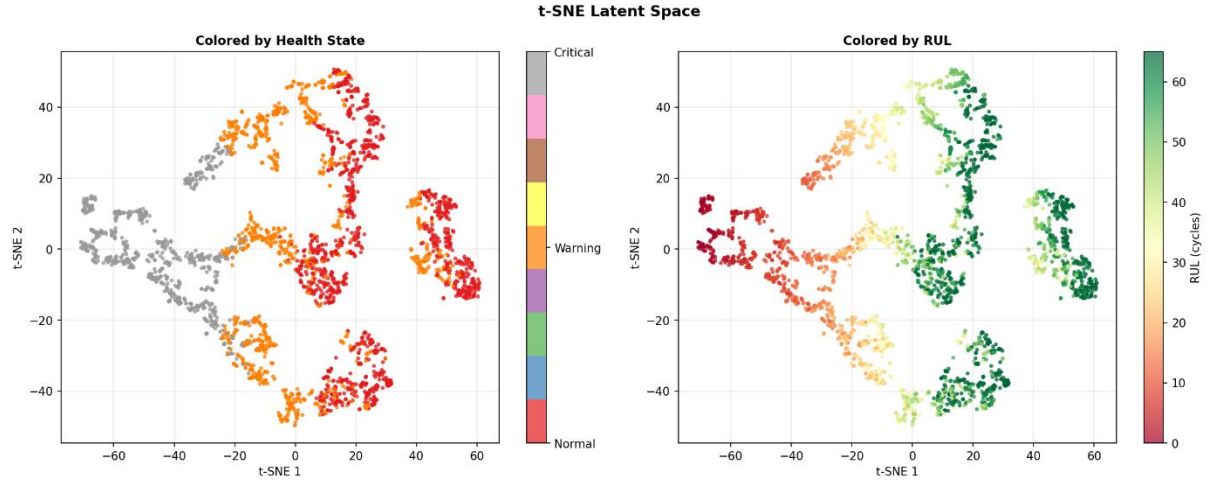


Figure 15. t-SNE latent space visualization of the proposed RG-GMamba framework.

The t-SNE based latent feature obtained through the proposed framework is shown in Figure 15. The proposed RG-GMamba model outperformed the traditional feature extraction methods, where the classes were clearly separated and had clear boundaries between normal, warning, and critical modes, while intra-class elements were grouped together in an optimized manner. Moreover, the latent representation based on the RUL shows gradual deterioration of performance with respect to time, where low remaining useful life data elements form clusters in more deteriorated regions of the feature space.

4.10 Feature Importance Analysis

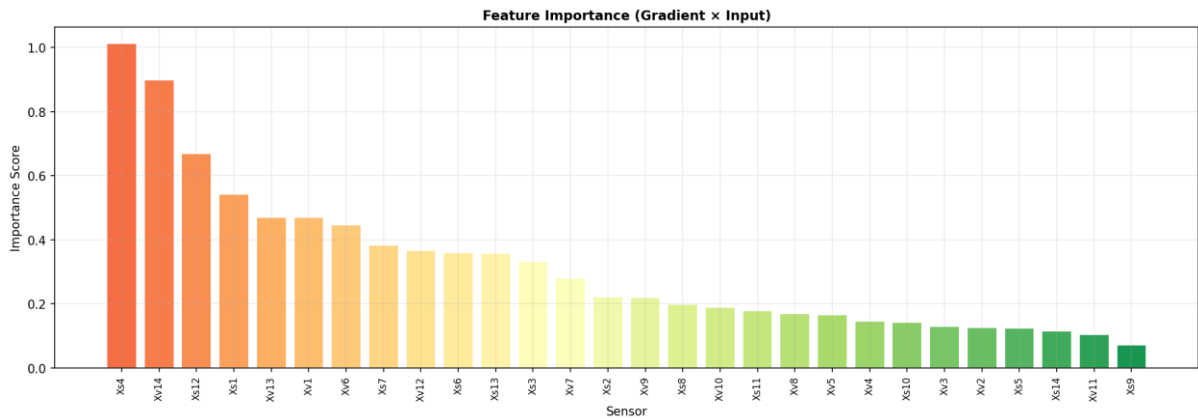


Figure 16. Gradient $\times$ Input feature importance analysis of the proposed RG-GMamba framework showing the contribution score of individual sensor variables.

Gradient $\times$ Input attribution technique was used to extract the sensor variable attributes which have the greatest impact on predicting Remaining Useful Life and health-state determination. The extracted distribution of feature importance is presented in Figure 16. From Figure 16, we observe that sensors Xd4, Xd14, and Xd12 attained the highest level of feature importance.

This means that they had great influence on degradation. The findings prove that the suggested RG-GMamba algorithm effectively filters out degradation sensitive sensor variables.

4.11 Explainable AI Analysis

To improve interpretability and transparency of the proposed RG-GMamba framework, explainable artificial intelligence techniques including SHAP and LIME are used. These methods helps to analyze the contribution of individual sensor variables toward Remaining Useful Life prediction and health-state classification decisions.

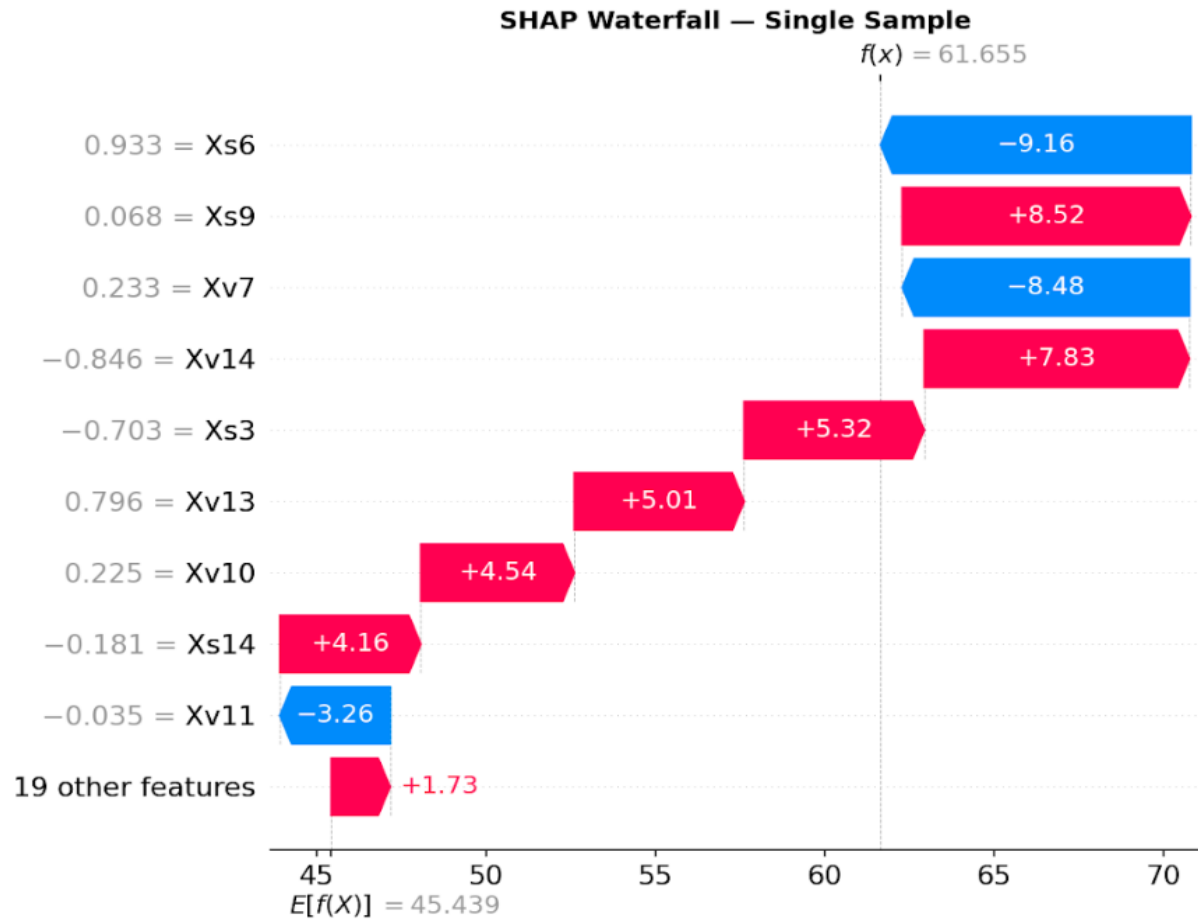


Figure 17. SHAP waterfall explanation of the proposed RG-GMamba framework showing positive and negative sensor contributions toward a single prediction output.

As shown in Figure 17, the SHAP waterfall chart shows how individual sensor-based variables affect the output prediction. Variables like Xs9, Xv14, and Xs3 have positively influenced the output prediction, while Xs6 and Xv7 have negatively influenced it. This indicates that the proposed model is able to detect both positive and negative impacts of degradation on various sensors.

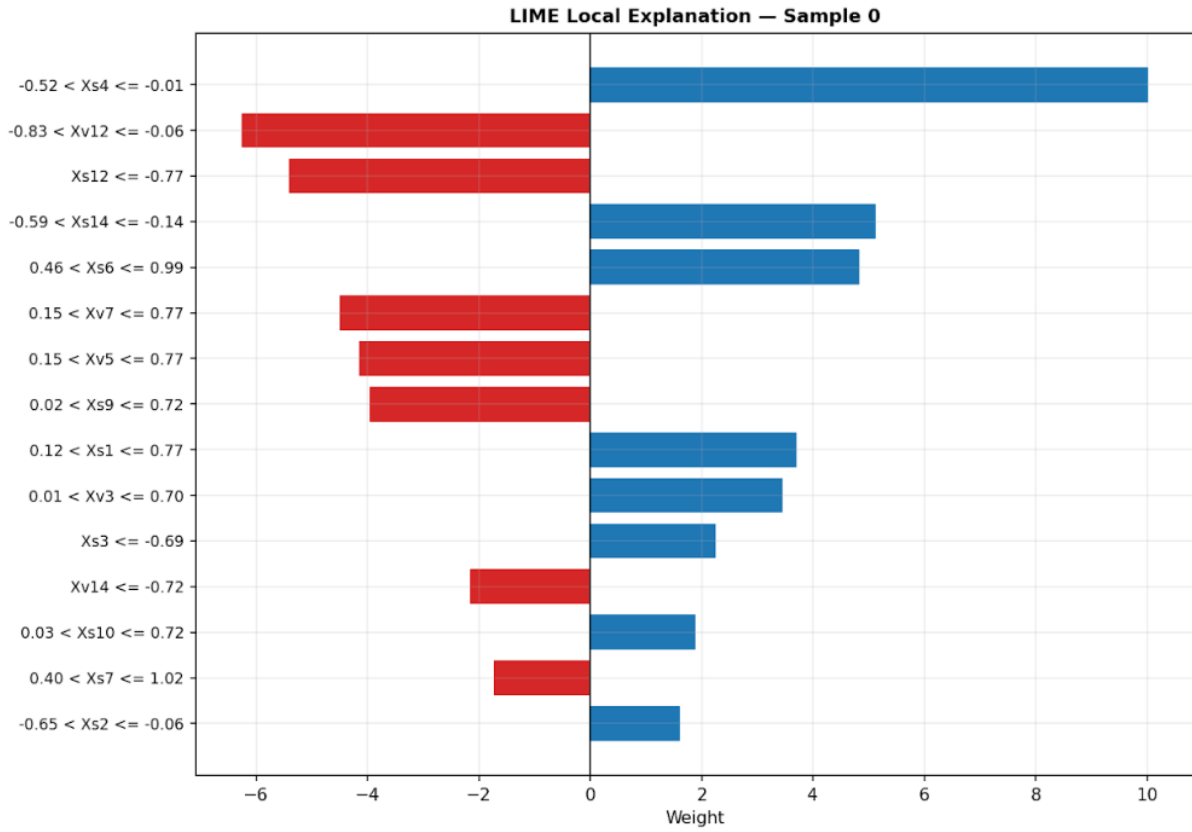


Figure 18. LIME-based local explanation analysis of the proposed RG-GMamba framework showing feature contributions for an individual prediction sample.

As shown in Figure 18 above, LIME-based local explanation is analyzed in the case of one sample prediction. Feature contribution having positive value is indicated on the right side, while those with negative values are illustrated on the left side. Features like X_{s4} , X_{s14} , and X_{s6} played an important role in predicting the condition of degradation state, while features such as X_{v12} and X_{v7} played a negative role in prediction. From the results above, the proposed RG-GMamba model offers explainability in decision making.

4.12 SOTA Model comparisons

As shown in Table 7, the proposed RG-GMamba framework achieved the best overall performance across both regression and classification tasks compared with existing predictive maintenance models. The proposed model obtained the lowest RMSE value of 6.3152 and MAE value of 4.8296, showing better Remaining Useful Life prediction accuracy and reduced degradation estimation error under different conditions. Furthermore, the R^2 score of 0.9058, shows strong correlation between predicted and actual Remaining Useful Life.

For health-state classification, the proposed RG-GMamba framework achieved the highest classification accuracy of 84.46% along with the best macro F1-score of 0.8484. This reflects a high ability of degradation state discrimination in case of Normal, Warning, and Critical states of operation. The proposed RG-GMamba framework demonstrated higher values of ROC-AUC of 0.9561 and PR-AUC of 0.9195 respectively.

Table 7. Comparison of Proposed RG-GMamba Framework with Existing Models

Model	RMSE ↓	MAE ↓	R ² ↑	Accuracy ↑	F1 Macro ↑	ROC- AUC ↑	PR- AUC ↑
LSTM-PdM	6.6840	4.9161	0.8944	0.8331	0.8361	0.9525	0.9112
CNN-BiLSTM	6.8959	5.0987	0.8876	0.8220	0.8256	0.9484	0.9004
Transformer-PN	6.4180	4.8417	0.9016	0.7533	0.7586	0.9531	0.9128
Proposed RG-GMamba	6.3152	4.8296	0.9058	0.8446	0.8484	0.9561	0.9195

As compared to the traditional recurrent network models like LSTM-PdM and CNN-BiLSTM, the proposed model was better because of its graph-based sensor interaction learning and state space temporal modeling, which helped to enhanced long-term degradation modeling capability along with efficiency. Moreover, in comparison to the Transformer-PN model, whose classification results were relatively weak despite its superior regression results, the proposed model had better multitasking performance regarding both the tasks.

4.13 Ablation study

Ablation experiments were carried out in order to measure the importance of each architectural element for the proposed RG-GMamba architecture. Different modules were gradually added, including the use of temporal convolutions, selective state space models, regime-aware gating mechanism, and learning of sensor interactions through graphs to analyze their impact on Remaining Useful Life prediction and health-state classification performance.

Table 8. Ablation Study of Proposed RG-GMamba Framework

Variant	Architecture	RMSE ↓	MAE ↓	R ² ↑	F1 Macro ↑	ROC- AUC ↑	NASA ↓
A1	CNN Only	10.6335	7.0150	0.7328	0.8115	0.9313	121864712.00
A2	CNN + Mamba	9.6106	7.3157	0.7818	0.7535	0.9050	50043.97
A3	W-Gate + CNN + Mamba	12.5747	9.8160	0.6264	0.6959	0.8470	88648.42
A4	W-Gate + CNN + GAT + Mamba	7.0488	5.1050	0.8826	0.8402	0.9485	28550.85
A5	Proposed Full RG-GMamba	6.3152	4.8296	0.9058	0.8484	0.9561	17630.09

From Table 8, it can be seen that while the CNN-only model (A1) performed reasonably well for predictive maintenance, it did not perform very well in modeling long-range degradation dependencies, resulting in high regression error and poor degradation modeling. Adding the Mamba selective state-space model (A2) was useful in enhancing temporal sequence learning by reducing RMSE from 10.6335 to 9.6106. However, classification accuracy was negatively impacted as no spatial inter-sensor dependency was modeled.

The introduction of the Regime-Aware W-Gate module in (A3) enabled the model to adapt sensor responses according to different operating conditions. While this mechanism helped reduce the influence of noisy sensor signals under varying flight regimes, the lack of explicit modeling of interactions among sensors limited the model's ability to fully capture degradation dynamics. As a result, the overall performance improvement remained modest. These findings suggest that regime-aware sensor modulation alone is not sufficient and should be complemented by effective inter-sensor relationship learning.

A notable improvement was achieved in (A4) with the inclusion of the Graph Attention Network (GAT) module. By learning the underlying thermodynamic and mechanical dependencies among sensor measurements, the model was able to better represent complex degradation patterns. Consequently, both regression and classification performance improved significantly, with the RMSE decreasing to 7.0488 and the macro F1-score increasing to 0.8402. This highlights the critical role of graph-based sensor interaction learning in enhancing aero-engine prognostics.

Finally, the complete RG-GMamba framework (A5) achieved the best overall performance across all regression and classification metrics. The integration of Regime-Aware Gating, Strided CNN-based temporal compression, Graph Attention learning, and Mamba-based selective state-space modeling enabled efficient spatiotemporal degradation representation learning, resulting in improved adaptability and computational efficiency.

5. Conclusion and Future Work

In this work, we proposed an AI-Enabled Predictive Maintenance Framework for Energy-Efficient IIoT Systems using a novel Regime-Gated Graph-Mamba (RG-GMamba) architecture for Remaining Useful Life prediction and health-state classification of aero-engines operating under varying industrial conditions. The proposed framework integrated regime-aware sensor modulation, lightweight temporal feature extraction, graph-based sensor interaction learning, selective state-space sequence modeling, and multitask learning to achieve accurate and energy-efficient predictive maintenance suitable for real-time IIoT environments. The framework successfully combined local degradation feature extraction, inter-sensor dependency learning, and long-range temporal degradation modeling within a unified architecture. Experimental evaluation on the NASA N-CMAPSS dataset demonstrated strong prognostic performance with an RMSE value of 6.3152, MAE of 4.8296, and R^2 score of 0.9058 for Remaining Useful Life prediction. The framework additionally achieved 84.46% classification accuracy, macro F1-score of 0.8484, ROC-AUC of 0.9561, and PR-AUC of 0.9195, while maintaining lightweight deployment capability with only 77,250 trainable parameters.

Future work will focus on Adaptive graph-topology learning and approaches such as Self-supervised learning or Contrastive Learning could be adopted for learning about degradation from labeled data available in an industrial setting. Multimodal monitoring in an industrial setting using vibration signals, audio, thermal imaging, and maintenance logs can also be explored to enhance degradation reasoning capability. The use of federated learning for distributed predictive maintenance in the IIoT network is also a promising approach.

6. References

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